

$$\begin{cases} \ln(y^2 - x) = 0 \\ x - y - 1 = 0 \end{cases} \cdot \begin{cases} y^2 - x = 1 \\ x = y + 1 \end{cases} \cdot y^2 - y - 2 = 0 \cdot \begin{cases} x_1 = 0 \\ y_1 = -1 \end{cases}, \begin{cases} x_2 = 3 \\ y_2 = 2 \end{cases}.$$

$$\begin{aligned} 1) \quad A_1 &= (0; -1). y = u - 1. \frac{d}{dt} \begin{pmatrix} x(t) \\ u(t) \end{pmatrix} = \begin{pmatrix} \ln((u-1)^2 - x) \\ x - u \end{pmatrix} = \begin{pmatrix} \ln(1 - 2u + u^2 - x) \\ x - u \end{pmatrix} = \\ &= \begin{pmatrix} -2u + u^2 - x + o(x) + o(u) \\ x - u \end{pmatrix} = \begin{pmatrix} -2u - x + o(\sqrt{x^2 + u^2}) \\ x - u \end{pmatrix} = \begin{pmatrix} -1 & -2 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ u \end{pmatrix} + \Psi. \end{aligned}$$

$(\lambda + 1)^2 + 2 = 0$. $\lambda^2 + 2\lambda + 3 = 0$. $D < 0$, $\text{Re}(\lambda) = -1$, точка равновесия устойчива.

$$\begin{aligned} 2) \quad A_2 &= (3; 2). \begin{cases} x = u + 3 \\ y = 2 + v \end{cases} \cdot \frac{d}{dt} \begin{pmatrix} u(t) \\ v(t) \end{pmatrix} = \begin{pmatrix} \ln((v+2)^2 - u - 3) \\ u + 3 - v - 2 - 1 \end{pmatrix} = \begin{pmatrix} \ln(1 + 4v + v^2 - u) \\ u - v \end{pmatrix} \\ &= \begin{pmatrix} 4v + v^2 - u + o(u) + o(v) \\ u - v \end{pmatrix} = \begin{pmatrix} 4v - u + o(\sqrt{u^2 + v^2}) \\ u - v \end{pmatrix} = \begin{pmatrix} -1 & 4 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} + \Psi. \end{aligned}$$

$(\lambda + 1)^2 - 4 = 0$. $\lambda_1 = -3$, $\lambda_2 = 1$, точка неустойчива.